Hysteresis, the Big Push, and Technological Adoption

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Bank of Finland Villa Research Workshop
August 17, 2022

Introduction

- Conventional view: GDP = Trend + Cycle
  - Trend= f(supply side factors)
  - Cycle= f(demand shocks and monetary policy).
  - Ideas $\rightarrow$ TFP.
- Alternative view
  - Relationship between the cycle and the trend.
  - Hysteresis: temporary shocks may have long run effects.
  - Ideas $\rightarrow$ implementation process $\rightarrow$ TFP.

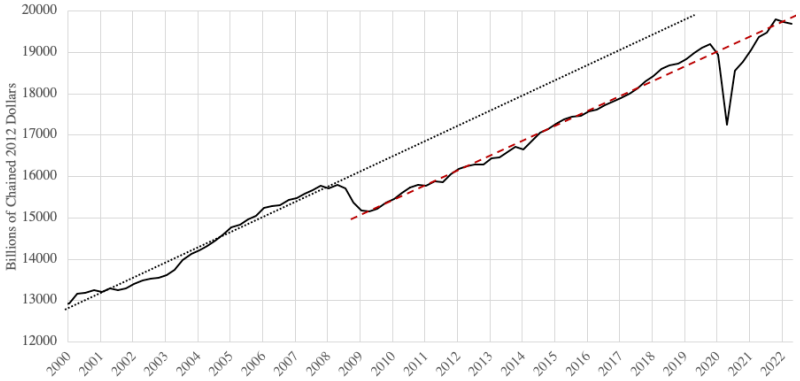


Figure: **United States Real GDP and Linear Trend (Source: FRED)**

This paper

- **Question:** which supply-side factors determine if a temporary shock has permanent effects?
- Focus on the **adoption** and **abandonment** of a technology by firms with heterogeneous costs.
- **Why not R&D?** supply-side scarring, relavance for non-R&D intensive economies.
- Coordination failure in the share of adopters (multiple equilibria, Big Push).
- Equilibrium selection: history of shocks $\rightarrow$ hysteresis may appear.
  - **Key feature:** if adoption threshold $>$ exit threshold $\rightarrow$ hysteresis.
  - Typical in stochastic settings when there is an option value of waiting for a high draw.
- Some results on the possibility of reverse hysteresis.

"In 1899 and 1900, electric vehicles outsold all other types of cars"

Figure: **Edison and electric car, 1913, (Source: National Museum of American History)**

Results

Hysteresis depends on:

- Elasticity of substitution between input varieties.
  - More sustituability/markups $\rightarrow$ more hysteresis.
- Barriers to adoption.
  - Higher entry costs $\rightarrow$ more hysteresis.
- Shock size.
  - Larger shocks $\rightarrow$ more hysteresis.

On the possibility of reverse hysteresis:

- Complementarity in adoption vs. decreasing profits for late adopters.
- It is relatively more expensive to adopt the higher the adoption share (holding costs constant).
- **Reverse hysteresis** more likely in less technologically advanced economies.
- **Friedman's (1964) plucking model** more likely in economies near/on the technological frontier.

Related Literature

- **Endogenous productivity, business cycles, and hysteresis:** Cerra, Fatás, and Saxena, forthcoming; Elfsbacka Schmöller, 2022; Garga & Singh, 2020; Anzoategui, Comin, Gertler, & Martinez, 2019; Bianchi, Kung, & Morales, 2019; Benigno & Fornaro, 2018; Aguiar & Gopinath, 2007;
- **Multiple equilibria based on capacity choice feedback:** Ramey, 1991; Doms & Dunne, 1998; Diamond, 1982; Cooper & Haltiwanger, 2006; Schaal & Taschereau-Dumouchel, 2018.
- **Technological choices and growth:** Hsieh & Klenow, 2014, Rosenstein-Rodan, 1945; Murphy et al., 1989; Ciccone, 2002; Comin & Hobijn 2010; Buera, Hopenhayn, Shin, & Trachter, 2021;

Contribution

- Alternative supply-side theory of hysteresis.
- Unlike previous work:
  - Not every shock has permanent effects.
  - Does not rely on R&D (more plausible for drops in trend and emerging economies).
- Explanation for the absence of reverse hysteresis (in technologically advanced economies).
- Explanation for Friedman's (1964) plucking model that does not rely on DNWR (e.g., Dupraz, Steinsson, and Nakamura, 2021).
- Alternative rationalization for trend shocks being more important in emerging economies (Aguiar and Gopinath, 2007).

Model

- Time is discrete and goes on forever.
- The economy:
  1. Representative household.
  2. Final good sector.
  3. Intermediate goods sector (continuum of varieties).
- Final good used for consumption and to pay for the cost of adopting and operating a modern technology.
- Resources used in adoption are "burnt" and are not used for any other purpose.

Model

Households:

- $\mathbb{E} \sum_{t=0}^{\infty} \rho^t U(C_t)$
- $U(C_t) = \ln C_t$
- $P_t C_t \leq W_t + \Pi_t$

Final goods producer:

- Production function: $Y_t = \left(\iint_{j,k} y_{j,k,t}^{\frac{\sigma-1}{\sigma}} \mathrm{d}j\mathrm{d}k \right)^{\frac{\sigma}{\sigma-1}}$
- Demand for variety j, k: $y_{j,k,t} = \left(\frac{p_{j,k,t}}{P_t} \right)^{-\sigma} Y_t$
- Final good price: $P_t = \left(\iint_{j,k} p_{j,k,t}^{1-\sigma} \mathrm{d}j\mathrm{d}k \right)^{\frac{1}{1-\sigma}}$

Intermediate goods producer

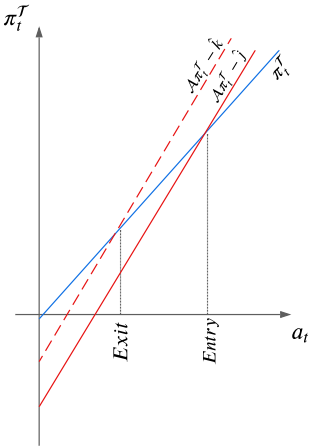
- Unit mass of firms indexed by adoption/abandonment thresholds (j, k).
- Varieties produced by monopolist choosing traditional $(\mathcal{T})$ or a modern $(\mathcal{M})$ technology.
- Heterogeneous entry $(\hat{j})$ and continuation $(\hat{k})$ costs at the firm level.
- Production function $y_{j,k,t} = A_{j,k,t} l_{j,k,t}$
- Productivity: $A_{j,k,t} = e^{a_t} u_{j,k,t}$
  - Shock: $a_t = \psi a_{t-1} + \varepsilon_t^a$,
  - If in the $\mathcal{T}$-sector, then $u_{j,k,t} = 1$
  - If in the $\mathcal{M}$-sector then $u_{j,k,t} = \mathcal{A} > 1$.

Intermediate goods producer

Adoption operator:

$$\gamma_{j,k,t} = \begin{cases} 1 & \text{if } \gamma_{j,k,t-1} = 0, \text{ and } \pi_t^{\mathcal{T}} > j \\ 1 & \text{if } \gamma_{j,k,t-1} = 1, \text{ and } \pi_t^{\mathcal{T}} > k \\ 0 & \text{if } \gamma_{j,k,t-1} = 0, \text{ and } \pi_t^{\mathcal{T}} < j \\ 0 & \text{if } \gamma_{j,k,t-1} = 1, \text{ and } \pi_t^{\mathcal{T}} < k \end{cases}$$

- Firm quantity and pricing decisions are independent of $\hat{j}$ and $\hat{k}$.
  - $\mathcal{T}$-firms: low quantities at high prices.
  - $\mathcal{M}$-firms: high quantities at low prices.



Intermediate goods producer

How the optimal thresholds are obtained for each firm?

- Assumption: $j \geq k$.
- Each firm forecasts $\pi_t^{\mathcal{T}}(a_t)$ and $m_t(a_t)$. The forecast density is obtained via simulation (no closed form due to truncation).
- Note that if $\pi_t^{\mathcal{T}} < \hat{j}$ profits will be negative if $\gamma_{j,k,t} = 1$.
- The value of the option to exit is computed as the expected loss (prob. $\times$ discounted loss). This corresponds to the actuarial fair premium of insuring negative profits. Denote this premium $\tilde{j}$.
- Then, $j = \hat{j} + \tilde{j}$.
- A similar logic can be applied to the exit threshold k.

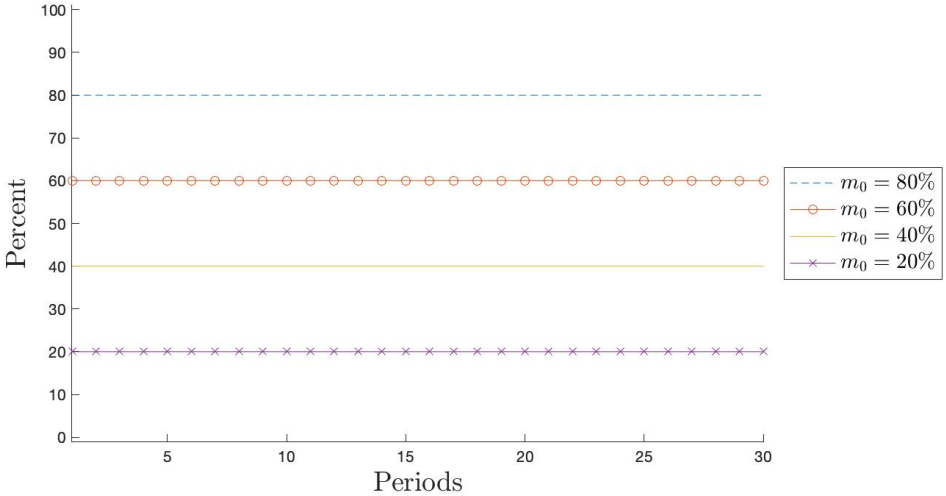
Coordination failures and multiple equilibria

- In equilibrium there will be complementarity in adoption:

$$\pi_t^{\mathcal{T}} = \frac{(\sigma-1)^{\sigma-1}}{\sigma^{\sigma}} \left(\frac{e^{a_t}}{W_t}\right)^{\sigma-1} Y_t$$

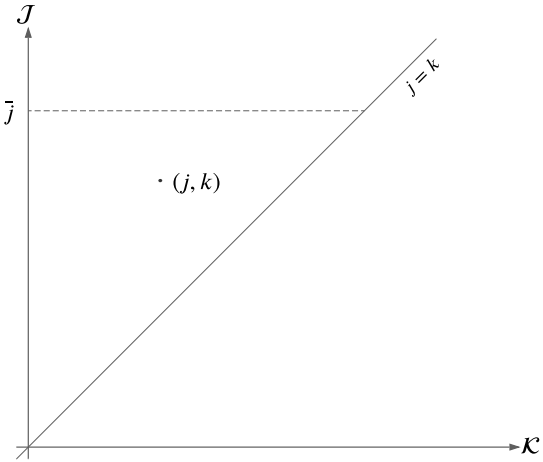
- Motive for coordination.
- Feedback from adoption to production decisions ($\gamma_{j,k,t}$).
- Result: multiple equilibria.

Continuum of steady state equilibria



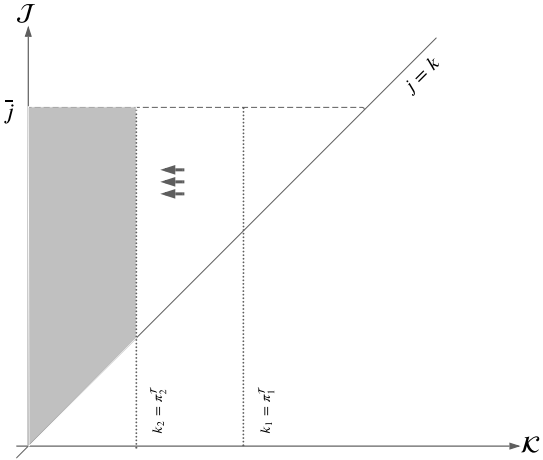
Equilibria selection based on history of shocks

Figure: **The $j \geq k$ half plane** $(t = 0)$



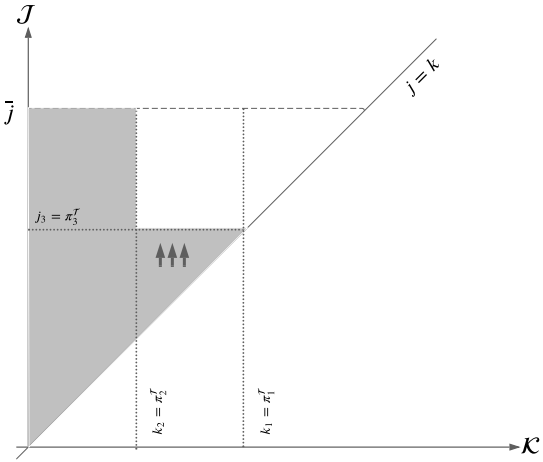
Equilibria selection based on history of shocks

Figure: $(t = 1)$



Equilibria selection based on history of shocks

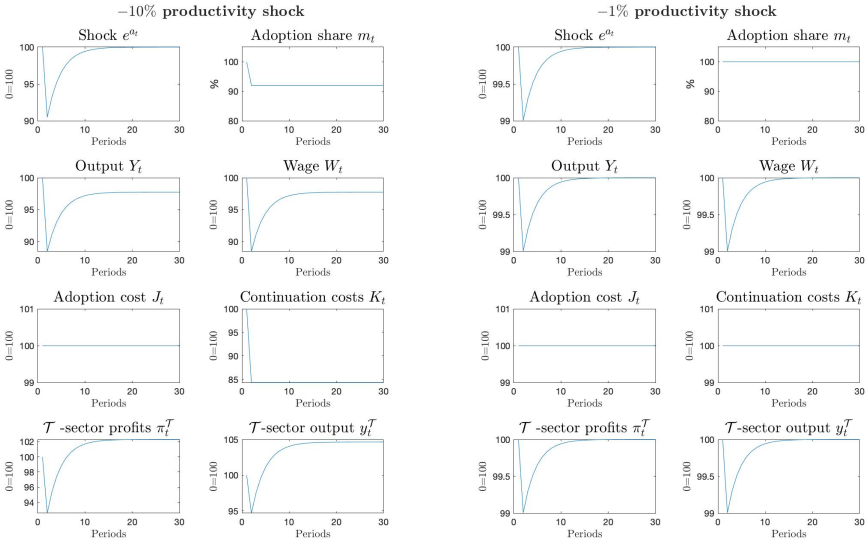
Figure: $(t = 2)$



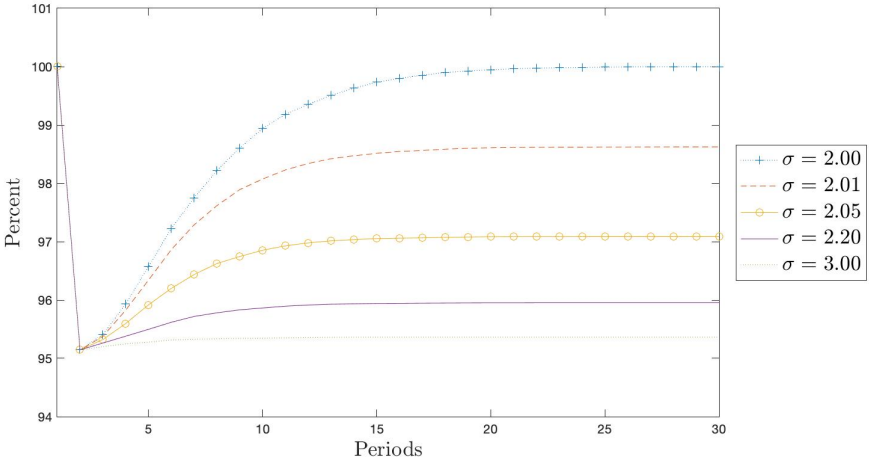
Computational exercises

1. Explore how different parameters affect hysteresis.
   - Shock size.
   - Market power.
     - Elasticity of substitution/markups.
     - Barriers to entry.
2. Reverse hysteresis and Friedman's plucking model.

Hysteresis depends on shock size

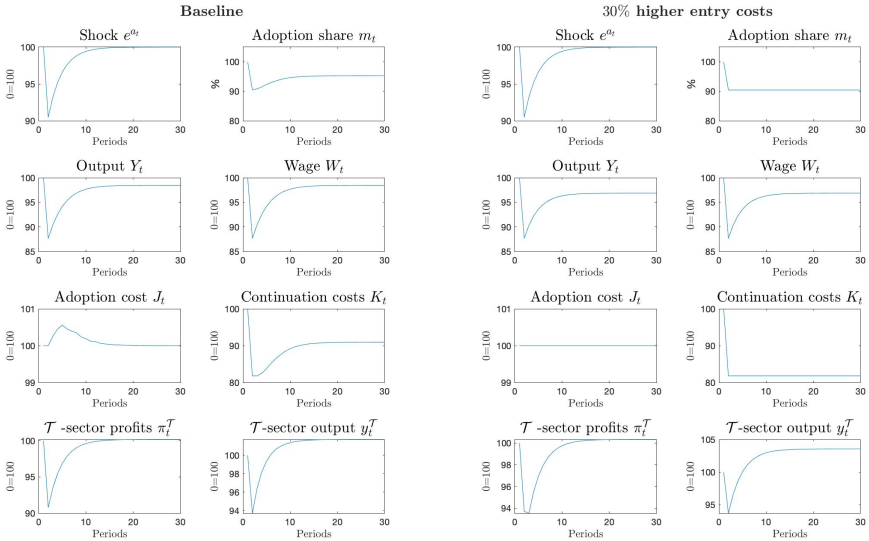


Hysteresis depends on markups ($\frac{\sigma}{\sigma-1}$)

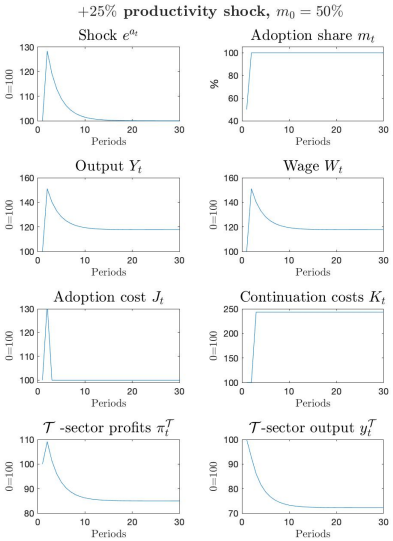


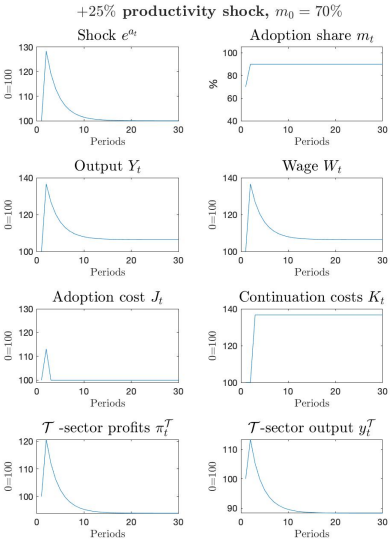
Note: These values of correspond to markups of 50 to 100%.

Hysteresis depends on entry costs j



Richer economies need bigger pushes





Conclusion

- Theoretical supply-side framework for hysteresis.
- Relevant for drops in trend and for non-R&D intensive economies.
- Defines a set of conditions for the possibility of reverse hysteresis.
- Novel equilibrium selection criterion based on history.
- Key mechanism based on heterogeneous adoption and abandonment thresholds.
- Results:
  - Hysteresis depends non-linearly on the size of the shock.
  - Hysteresis depends on market power (markups, entry costs).
  - Reverse hysteresis more likely in economies far away from the technological frontier.
  - Friedman's plucking model more likely for economies on the technological frontier.